



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

EXPERIMENT :2

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Section/Group: KRG-1A

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Subject Name: ADBMS

Subject Code: 23CSP-333

1.Aim

Medium-Level Problem

Problem Title: You are a Database Engineer at TalentTree Inc., an enterprise HR analytics platform that stores employee data, including their reporting relationships. The company maintains a centralized Employee relation that holds: Each employee's ID, name, department, and manager ID (who is also an employee in the same table).

Your task is to generate a report that maps employees to their respective managers, showing:

- The employee's name and department.
- Their manager's name and department (if applicable).
- . This will help the HR department visualize the internal reporting hierarchy.

Hard-Level Problem

Problem Title: To write SQL queries that retrieve requested Net Present Value (NPV) data by performing conditional joins on multiple columns (ID and YEAR) across two related tables, and to handle missing data using appropriate SQL functions.

2. Tools Used: SQL Server Management Studio

3. Query

Medium Level Problem

```
create table employee_tbl(emp_id integer primary key, emp_name varchar(50) NOT NULL, dept  
varchar(50) NOT NULL, manager_id int NULL );
```

```
alter table employee_tbl
```

```
add constraint fk_manager FOREIGN KEY(manager_id) REFERENCES employee_tbl(emp_id);
```

```
insert into employee_tbl(emp_id,emp_name, dept, manager_id) values(1,'Alice', 'HR',null);
insert into employee_tbl(emp_id,emp_name, dept, manager_id) values(2,'Bob', 'Finance',1);
insert into employee_tbl(emp_id,emp_name, dept, manager_id) values(3,'Jay', 'IT',1);
insert into employee_tbl(emp_id,emp_name, dept, manager_id) values(4,'William', 'IT',3);
insert into employee_tbl(emp_id,emp_name, dept, manager_id) values(5,'Tony', 'Finance',2);
select * from employee_tbl;
```

```
select E1.emp_name, E1.dept , E2.emp_id as [manager_id], E2.dept as [manager_dept]
FROM employee_tbl as E1
INNER JOIN
employee_tbl as E2
ON E1.manager_id=E2.emp_id;
```

Hard Level Problem

```
create table year_tbl(id integer , year int ,npv int );
insert into year_tbl(id, year, npv) values(1, 2018,100);
insert into year_tbl(id, year, npv) values(7, 2020,30);
insert into year_tbl(id, year, npv) values(13, 2019,40);
insert into year_tbl(id, year, npv) values(1, 2019,113);
insert into year_tbl(id, year, npv) values(2, 2008,121);
insert into year_tbl(id, year, npv) values(3, 2009,99);
insert into year_tbl(id, year, npv) values(11, 2020,12);
insert into year_tbl(id, year, npv) values(7, 2019,0);
```

```
create table query_tbl(id integer , year int );
insert into query_tbl(id, year) values(1,2019);
insert into query_tbl(id, year) values(2,2008);
insert into query_tbl(id, year) values(3,2009);
insert into query_tbl(id, year) values(7,2018);
insert into query_tbl(id, year) values(7,2019);
```

```
insert into query_tbl(id, year) values(7,2020);
```

```
insert into query_tbl(id, year) values(13,2019);
```

```
select * from year_tbl;
```

```
select * from query_tbl;
```

```
select Q.id, Q.year, ISNULL(Y.npv, 0) as npv
```

```
from year_tbl as Y
```

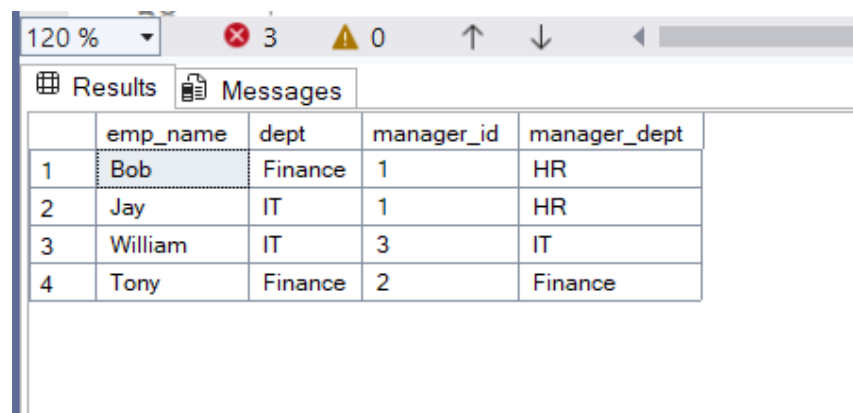
```
RIGHT JOIN
```

```
query_tbl as Q
```

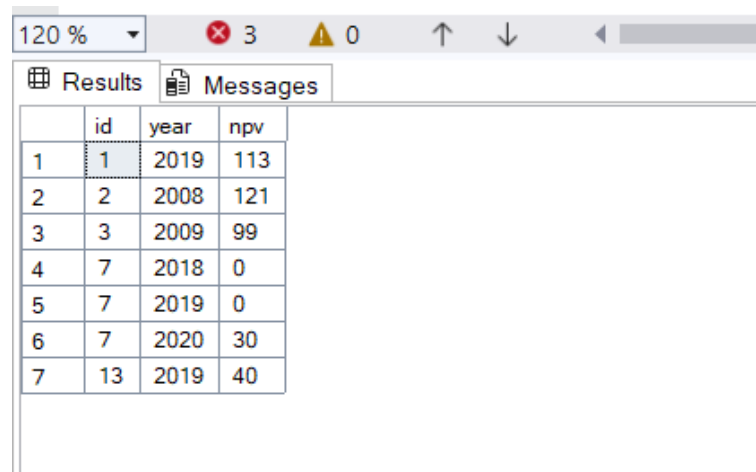
```
ON
```

```
Y.id= Q.id and Y.year = Q.year;
```

4. Output:



	emp_name	dept	manager_id	manager_dept
1	Bob	Finance	1	HR
2	Jay	IT	1	HR
3	William	IT	3	IT
4	Tony	Finance	2	Finance



	id	year	npv
1	1	2019	113
2	2	2008	121
3	3	2009	99
4	7	2018	0
5	7	2019	0
6	7	2020	30
7	13	2019	40



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5. Learning Outcomes

- a) Learn to implement self-joins to represent hierarchical relationships within a single table (e.g., employees reporting to managers).
- b) Construct relational queries to retrieve information such as employee–manager relationships, handling missing data with LEFT JOIN.
- c) Use functions like ISNULL() to replace missing values with defaults during join operations.
- d) Understand how to design normalized relational tables with foreign key constraints for real world entities like departments and courses.